DLMI2026

Hands-On: Vision Foundation Models for Computational Pathology

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ÉTS Le génie pour l'industrie

cnrs

itech santé

CIMS Centre d'imagerie moléculaire de Sherbrooke

CHUS Centre hospitalier universitaire de Sherbrooke

Fonds de recherche Nature et technologies Québec

LABEX PRIMES UNIVERSITÉ DE LYON

MICCAI

CREATIS

UdS Université de Sherbrooke

Objectives of the Hands-On

- Understand pathology image representations
- Learn how foundation models are trained
- Extract embeddings using PLIP
- Perform zero-shot classification
- Compare zero-shot and supervised learning
- Train a simple classifier on extracted features

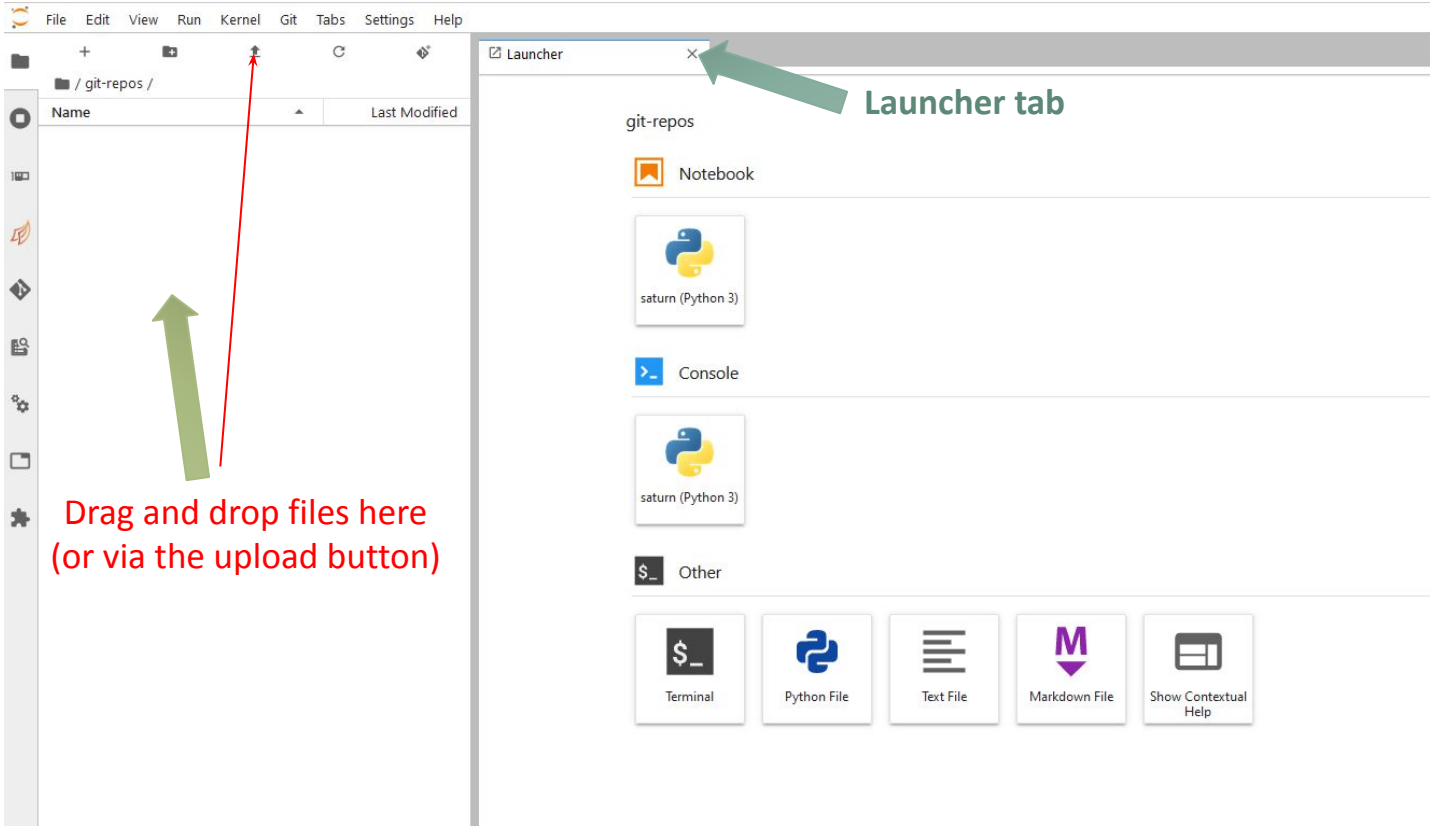
Steps required

1. *Have an account on CalculQuebec Cloud and log-in*
   https://jupyter.dmli.calculquebec.cloud/hub/login
2. Create a compute node with
   1. Memory at 4096MB
   2. GPU 1x1G.5GB

Steps covered here

3. Start the jupyter-lab server and wait for it
4. Download the hands-on material into your jupyter-lab
5. Open a notebook
6. Cells philosophy and execution

3- Jupyter Lab interface (can change)



4- Download Hands-on materials : **2 ways**

Way « on server »

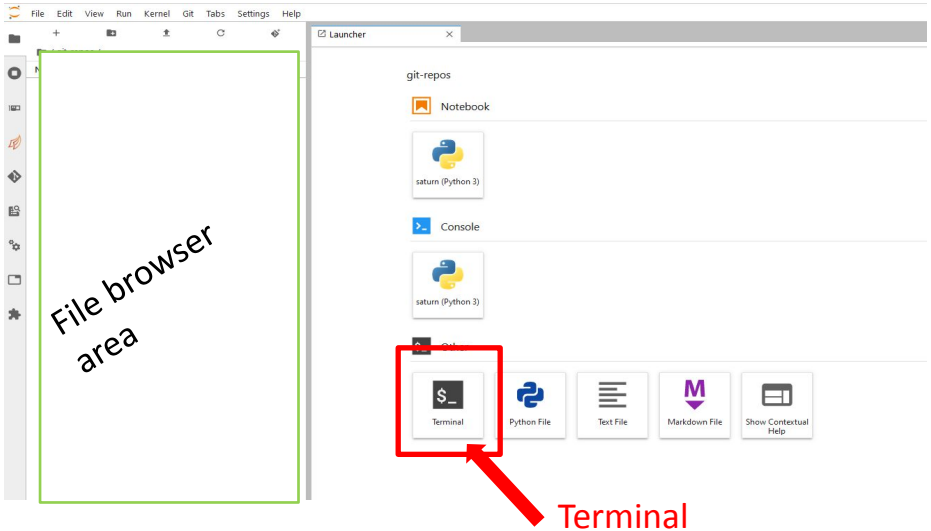
a- Open a Terminal

b- download from the terminal with :

```
git clone
https://github.com/omchakrabarty/DIML_SummerSchool.git
```

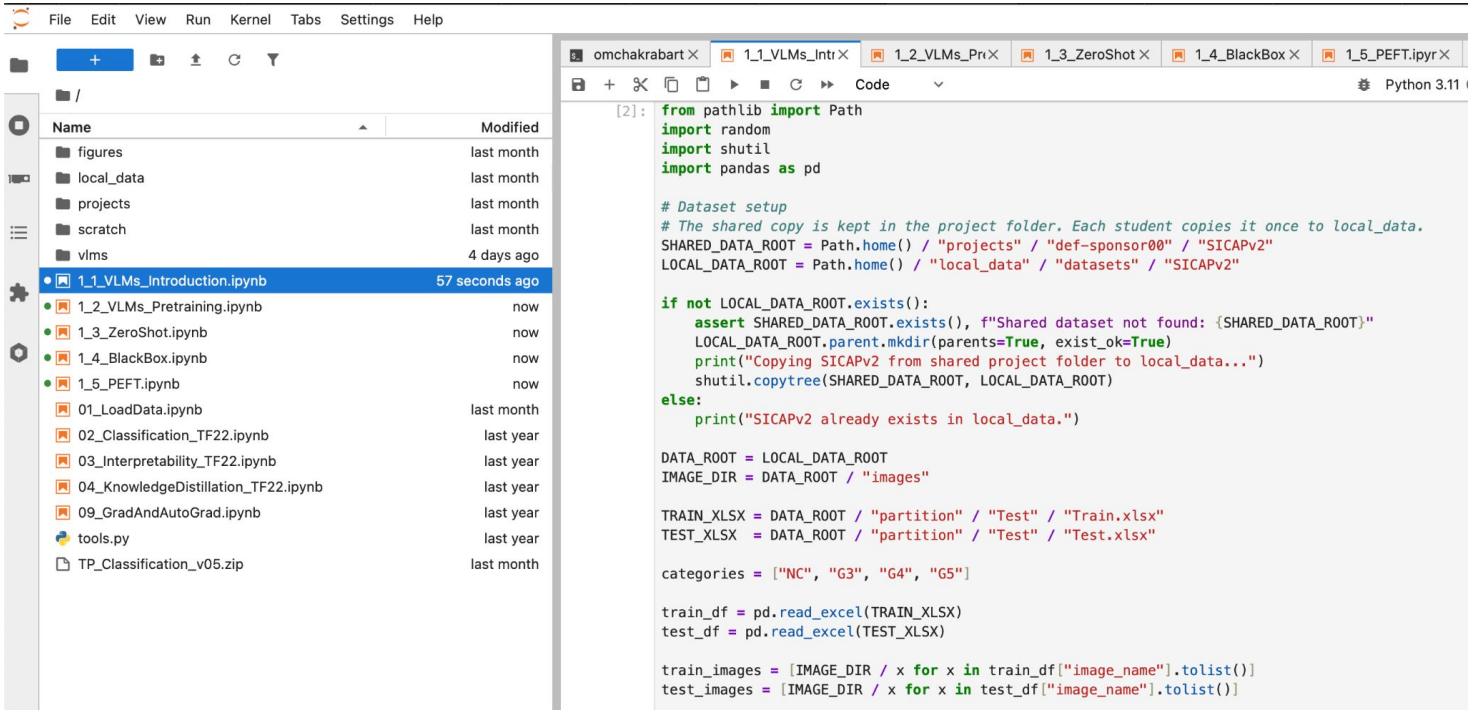
Way « Drag and drop » (recommended)

a- download on your computer the hands-on material as zip file (links? : next page!)
b- drag on drop it in the **file browser area**
c- unzip from jupyter lab (in a Terminal)



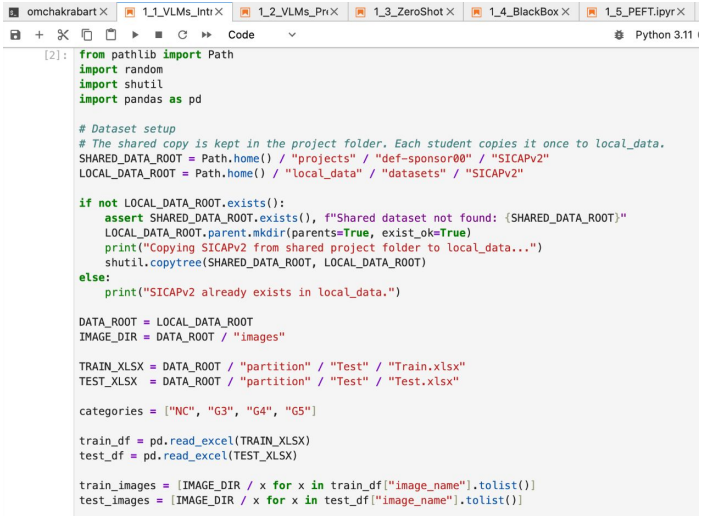
5- Select and open the desired notebook

As an example, double click on : 1_1_VLMs_Introduction.ipynb



6- Cells and running code

- Notebooks are made of cells : they can be text or python code
- Python code can be executed from the jupyter lab with play button
- Outputs are added in a new cell after the code executed
- A cell can be executed many times
- Code of a cell can be changed
- Generally, a cell requires that the preceding cells have been executed before



7- Troubleshooting

You can find a troubleshooting section at the very end of the Github's README with common problems you might encounter. Check there first in case need.